Supplementary Information

for the article entitled *"Universality of the fluorescence intermittency in nanoscale systems: experiment and theory"* by Pavel A. Frantsuzov, Sándor Volkán-Kacsó and Boldizsár Jankó.

Contents

I. Power spectral density estimation from experimental trajectories 1

A. Point estimation of the power spectral density 1

B. Estimation of confidence intervals 4

II. Estimation of the power law exponents and switching rates 8

III. Additional details of the MRC framework 9

IV. Calculating the power spectral density in the MRC model 12

V. System-specific statistical features of fluorophore trajectories 18

References 20

I. POWER SPECTRAL DENSITY ESTIMATION FROM EXPERIMENTAL TRAJECTORIES

The purpose of this section is to provide a detailed description of the spectral estimation method used for all seven blinking fluorophore systems discussed in the main article. In particular, we address two issues, (1) we describe the point estimator for the power spectral density (PSD), and (2) we explain how we compute the confidence intervals. Specific details for each trajectory are given in the final paragraph of the SI.

A. Point estimation of the power spectral density

We follow the standard PSD estimation method by Welch, the modified average periodogram (MAP) estimator [1]. The method was originally introduced for discrete time, an example of which is binned intensity fluctuation trajectory. This method prescribes (1) the partitioning (or windowing) of the time series $\{x_1, x_2, ..., x_N\}$ into a number K of smaller blocks containing M

elements, (2) the calculation of the modified periodograms of each block and (3) the averaging of the resulting periodograms. Given a (binning) time unit of τ_0, we first define the periodogram of the whole time series, the simplest PSD estimator, as [2]

$$\hat{S}_{\rm per}(f) = \frac{\tau_0}{N}\left|\sum_{n=1}^{N} x_n \exp\{i2\pi f\tau_0(n-1)\}\right|^2, \tag{S1}$$

where the frequency takes discrete values $f = k/(\tau_0 N)$, $k = 0, 1, ..., N-1$. It can be shown for the simple periodogram in Eq. (S1) that the standard deviation follows the relation $\hat{\sigma}_{\rm per} \sim \hat{S}_{\rm per}$ independent of how large N is, rendering $\hat{S}_{\rm per}$ an inconsistent estimator [2]. A modified periodogram results from applying some specially designed non-uniform weight function in the summation. This discrete function will be called window function since it can also serve to partition the original time series. The windowing can be both overlapping or non-overlapping, and is obtained by convolving the trajectory with the window function w_n. For non-overlapping windows $K = [N/M]$, here the $[\cdot]$ operator stands for integer part (floor), and the resulting blocks are of the form $w_n x_{(mM-n)}$. The modified periodogram estimations for the mth block thus can be formally given as

$$\hat{S}_m(f) = \frac{\tau_0}{M}\left|\sum_{n=1}^{M} w_n x_{(mM-n+1)} \exp\{i2\pi f\tau_0(n-1)\}\right|^2, \quad m = 1, 2, ..., K. \tag{S2}$$

The generalization to overlapping windows is straightforward. Finally we arrive to the definition of the MAP estimator as

$$\hat{S}_{\rm MAP}(f) = \frac{1}{K}\sum_{m=1}^{K} \hat{S}_m(f). \tag{S3}$$

The averaging is needed in order to reduce the variance of the periodogram estimator.

The window functions assign more weight to points in the center then to peripheral points in order to reduce the bias resulting from finite size effect. In practice one of the most successful designs is the Hamming window, defined in the present notation as

$$w_n = \frac{1}{2} + \frac{1}{2}\cos\pi[(n-1)/M' - 1/2], \quad n = 1, 2, ..., M, \tag{S4}$$

where $M' = (M-1)/2$ and M must be odd. Independent of the choice of the window [28] the bias is finite and can be systematically reduced by increasing the window size. Given only a single realization of the stochastic blinking process per fluorophore we are lead to the classical bias versus variance trade-off problem: longer window means smaller bias, but also larger variance due to fewer

independent estimates.

In lieu of an *a priori* model for the PSD (as mentioned, we use a nonparametric method), we have no simple estimation available for the size of the bias induced by the finiteness of the trajectory. Instead we choose to remove the bias by the following method [29]. In order to achieve low variances, we choose the shortest possible window size that still shows negligible bias. That can be done by inspecting the window size dependence of $\hat{S}_{\mathrm{MAP}}$. This procedure is applied for each point, so the window size scales with the frequency. Fig. S1 shows an example of a series of windows and the corresponding frequencies. The colored contour lines indicate the estimated values of the PSD as a function of the two parameters: window size and frequency. Clearly, the presence of straight vertical contour lines indicate a region of no window size dependence, a signature of negligible bias. The window-frequency pairs must be within this no-bias region, but also close to small widow size, for minimum variance.

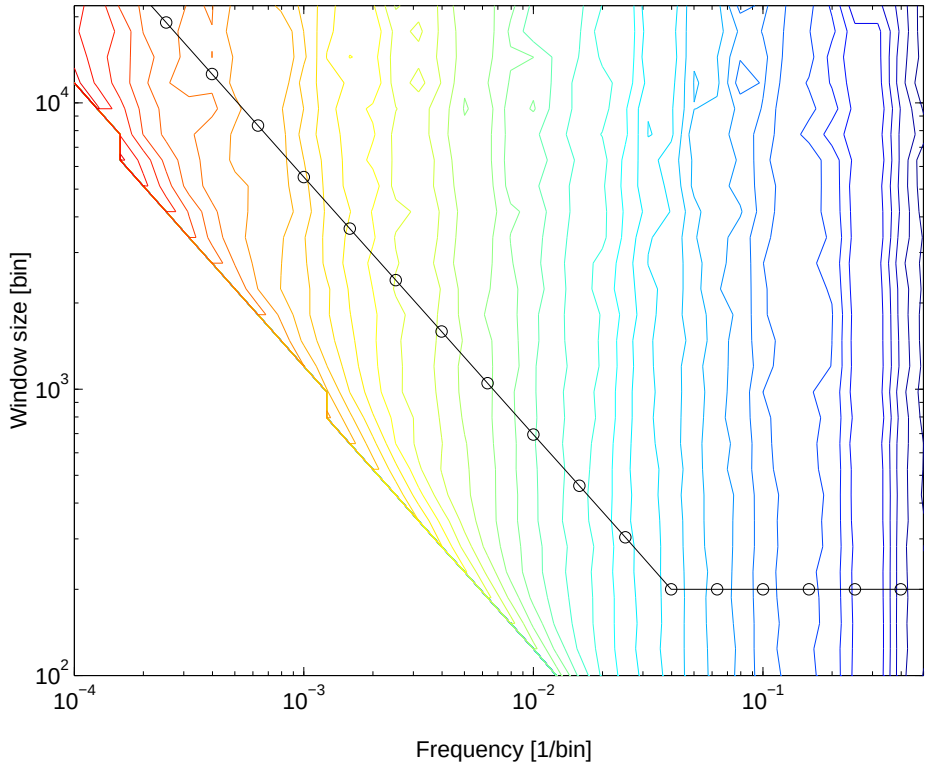


FIG. S1: The average periodogram estimation of the PSD of the CdSe core-shell quantum dot trajectory of Fig. 2 for different window sizes. The contours correspond to lines of identical PSD estimates. The colors identify different values of the PSD estimates. The connected black circles show the chosen frequency-window series of the estimations used in Fig. 2.II. The PSD was not calculated in the white field of lower-left where window sizes are comparable or smaller then the inverse of the estimation frequencies.

In practice the fluorescence fluctuation from most single fluorophores has times scales that vary over several orders of magnitude. Thus we typically sample the power spectrum uniformly on a

logarithmic scale. We then associate to these frequencies a window size according to the principle of minimum variance and negligible bias. From Fig. S1 clearly follows that it is necessary to scale window sizes with inverse frequency. We make an exception for high frequencies, where a constant window size is chosen. This is needed in order to minimize the bias due to the discreteness of the trajectory. For example, the CdSe-ZnS core-shell dot trajectory is long enough so that we can keep the minimum window size at 200 bins and still get reasonable variance reduction. The frequency-window pairs for the trajectory of Fig. 2.II is shown as an example in Fig. S1, while in Fig. S2 these pairs were plotted for all trajectories analyzed in this work.

In conclusion, our method is a variable window version of the classical MAP estimator and shows similarities to other, less standard methods of spectral estimation [3]. While in the MAP method the window size is usually a free parameter, we have given here a prescription to choose an optimal window size.

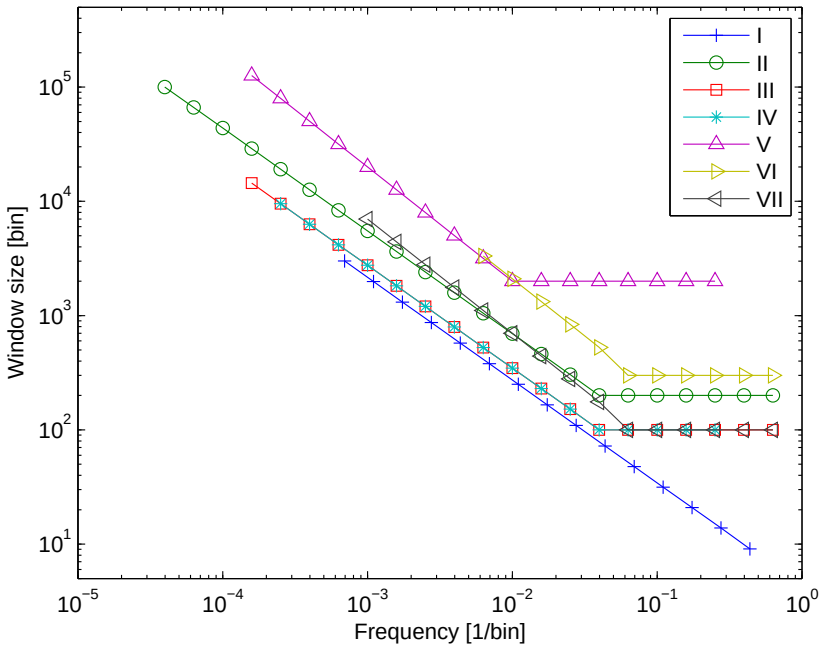


FIG. S2: Window sizes used for the spectral estimation of Fig. 2 for all trajectories (I-VIII), plotted as a function of frequency values in natural units (time bins).

B. Estimation of confidence intervals

The number of data available for the different trajectories of Fig. 2 and their spectral property greatly varies. For instance, the length of CdSe/ZnS trajectories (Fig 2.II) is 360000 time bin units,

while those of the quantum wires (Fig. 2.VI) are only about 6500 units. Also, some trajectories fluctuate on only one time scale (the PSD can be fitted by a Lorentzian), while others on a range of timescales varying over almost 5 decades. Nevertheless, we strived to develop an error estimation method that is applicable to all trajectories. Due to the greatly varying number K of windows (and estimates $\{\hat{S}_m(f)\}$) we are in one of two possible regimes. In the first regime, which applies to the great majority of frequencies, $K \gg 1$ and one can calculate a significant population of estimates $\{\hat{S}_m(f)\}$ from windowing. In this regime we apply a generalized version of the central limit theorem (CLT) to argue that $\hat{S}_{\rm MAP}(f)$ follows a Gaussian distribution [4]. The second regime sets in when only a reduced number (perhaps $K \sim 10$) of such estimates are available and the Gaussian approximation is too crude. Note that this regime applies for the very low end of the frequencies, which in practice means no more then a half decade in frequency domain. In other words, only the first 3-4 points of our spectral estimation fall in this regime.

As mentioned, for most of the frequencies the first regime applies, meaning that the number of estimates falls roughly in the interval $20 - 2000$. With such moderate to large sample sizes we are aiming to use the CLT theorem. In its simplest form the CLT theorem states that the average of a large sample drawn from some distribution tends to be distributed according to a Gaussian PDF. Two conditions must apply: (1) that the estimates from the sample must be statistically independent, and (2) the population variance must be well defined. In the most convenient case of the $\{\hat{S}_m(f)\}$ being statistically independent, $\hat{S}_{\rm MAP}(f)$ is normally distributed with standard deviation $\mathrm{std}[\hat{S}_{\rm MAP}(f)] = \sqrt{\left\langle\left\langle \hat{S}_m(f)^2 \right\rangle\right\rangle / K}$. In our error estimation we choose the standard 95% confidence interval, which corresponds to an error bar that extends $\pm 1.96\ \mathrm{std}[\hat{S}_{\rm MAP}(f)]$ relative to the value of the PSD estimation $\hat{S}_{\rm MAP}$ [5]. We can find statistically independent $\{\hat{S}_m(f)\}$ in some cases, eg. trajectories from Fig. 2.I and 2.IV.

Typical for our PSD estimation procedure, a relatively small window size is associated with a large number of estimates. Due to the long range correlations present in the 1/f trajectories of Fig. 2 (eg. II, III, IV, VI and VII), the periodogram estimates from neighboring windows are correlated. While the condition for statistical independence of $\{\hat{S}_m(f)\}$ can be relaxed for weak dependence [6], neighboring estimates on small window sizes are clearly correlated. We therefore used an extended version of CLT, which takes into account the statistical dependence between estimates [4]. This theorem applies if the covariance $\kappa_m(f) = \mathrm{cov}\{S_n(f), S_{n+m}(f)\}$ becomes effectively negligible for some $m > m_c$.

Our implicit assumption is that the time-ordered series (chain) $S_1(f), ..., S_K(f)$ is stationary.

Then the variance of the mean can be shown [1] to be

$$\kappa_{\mathrm{MAP}}(f) = \frac{1}{K}\left[\kappa_0(f) + 2\sum_{m=1}^{m_c}\left(1-\frac{m}{K}\right)\ \kappa_m(f)\right]. \tag{S5}$$

The variance $\kappa_0(f)$ and covariances $\kappa_m(f)$ can be estimated as [1]

$$\hat{\kappa}_m(f) = \frac{1}{K-m}\sum_{n=1}^{K-m}\hat{S}_n(f)\hat{S}_{n+m}(f). \tag{S6}$$

In practice we investigate the validity of approximate statistical independence beyond a separation of m_c by evaluating the cumulative quantity

$$\hat{\kappa}_k^{\mathrm{cum}}(f) = (1/K)[\hat{\kappa}_0(f) + 2\sum_{m=1}^{k}(1-m/K)\hat{\kappa}_m(f)] \tag{S7}$$

as a function of k. It is expected that $\hat{\kappa}_k^{\mathrm{cum}}(f)$ is a monotonous function of k that does not increase beyond some 'saturation' point, which we can identify with m_c. Another assumption is that higher order correlations decay at least as fast as $\kappa_m(f)$. In our procedure we choose the maxima of $\hat{\kappa}_k^{\mathrm{cum}}(f)$ as our estimation for the variance of the mean,

$$\hat{\kappa}_{\mathrm{MAP}}(f) = \max_k\{\hat{\kappa}_k^{\mathrm{cum}}(f)\} \tag{S8}$$

and the maximum point as the estimation for m_c. It is important to note that in many cases there is a significant fluctuation of $\hat{\kappa}_k^{\mathrm{cum}}(f)$ as k approaches K (Fig. S3). Thus by choosing the maximum it is likely that we somewhat overestimate the resulting confidence interval. We nevertheless chose to follow this conservative estimate for the error bars. The 95% confidence interval corresponds to an error bar defined between points $\hat{S}_{\mathrm{MAP}}(f) \pm 1.96\ \sqrt{\hat{\kappa}_{\mathrm{MAP}}(f)}$.

We also investigated the nature of the probability distribution of $\{\hat{S}_m(f)\}$ which defines the behavior of the variance associated to $\hat{S}_{\mathrm{MAP}}(f)$. For each point in the estimated spectra we performed an empirical test based on evaluating the rank-ordering statistics of $\{\hat{S}_m(f)\}$, an estimation of the cumulative distribution [7]. We noticed that the distribution typically has a maximum close to zero and a tail always decaying faster than $1/x^3$. Consequently CLT can be safely applied to the data and $\hat{\sigma}_{\mathrm{MAP}}(f) \to 0$ as K increases (ref. [7]).

In order to validate our estimator $\hat{\kappa}_{\mathrm{MAP}}(f)$ and the Gaussian nature of the distribution of $\hat{S}_{\mathrm{MAP}}(f)$ we also implemented resampling schemes, such as the moving block bootstrap (MBB) [8] and the circular block bootstrap (CBB) [9]. Both MBB and CBB yielded variances that are in

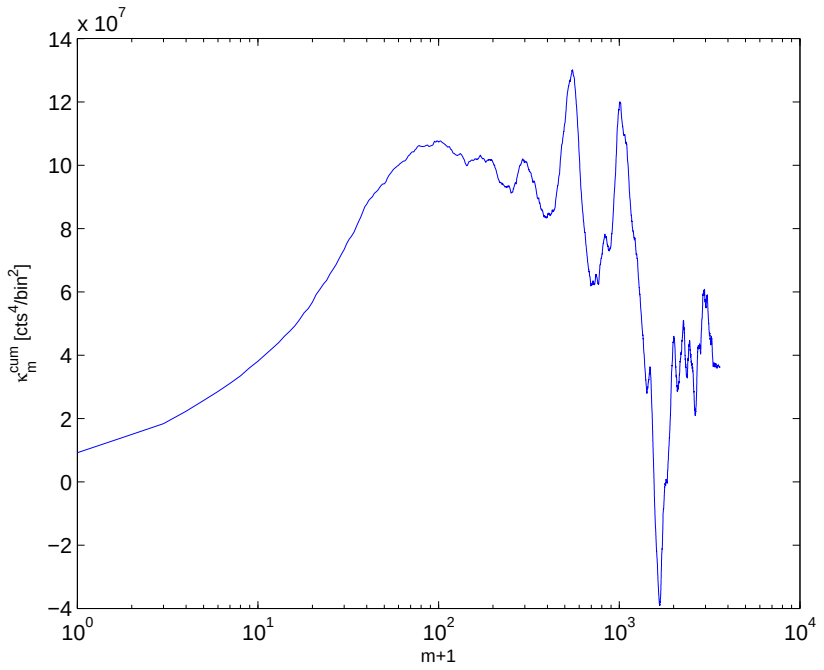


FIG. S3: An example of the estimated cumulative variance $\hat{\kappa}_k^{\rm cum}(f)$ as a function of the window index variable k. The trajectory is that of Fig. 2.II and the frequency of the PSD estimation is f=6.3Hz. The $\hat{\kappa}_k^{\rm cum}$ reaches the maximum at $m_c \sim 100$

good agreement with $\hat{\kappa}_{\rm MAP}(f)$.

We now turn to the second regime we typically find in our PSD estimation problem, which is in the limit of low frequencies and correspondingly long but few ($\sim$ 10) windows. In most cases one finds that the covariances are negligible, $\hat{\kappa}_0 >> 2\hat{\kappa}_1$, meaning that the set $\{\hat{S}_m\}$ can be regarded as essentially uncorrelated. Assuming, as we did earlier, that higher order correlations are unimportant, this further translates into statistical independence. Thus, provided a large enough K and the fact that the CLT applies, the 95% confidence intervals can be easily given as $\hat{S}_{\rm MAP} \pm 1.96\sqrt{\hat{\kappa}_0/K}$. However, if we only posses a few independent estimations, the Gaussian nature of the distribution of mean might be a crude approximation. We can, however, deal with this situation by applying a well-established method [2]. It turns out, that the $\{\hat{S}_m\}$ sets closely follow a simple exponential PDF,

$$p(\hat{S}_m) = \frac{1}{S}\exp\left(-\frac{\hat{S}_m}{S}\right). \tag{S9}$$

where S is PSD expectation value. The mean of K independent estimators $\hat{S}_{\rm MAP}$ Eq. (S3) follows the well known χ^2 distribution with $2K$ degrees of freedom. Let us introduce new variable $R =$

$\chi^2 = 2K\hat{S}_{\rm MAP}/S$. The distribution of R follows [5].

$$p(R) = \frac{R^{K-1}}{2^K \Gamma(K)} \exp\left(-\frac{R}{2}\right), \tag{S10}$$

Note that the PDF in Eq. (S9) is the χ^2 distribution with 2 degrees of freedom. We denote the percentage points 2.5% and 97.5% with χ^2_1 and respectively χ^2_2. These values can be calculated from the inverse of the χ^2 cumulative distribution function. Then the 95% confidence interval for S value is given by the following numbers $(2K\hat{S}_{\rm MAP}/\chi^2_2, 2K\hat{S}_{\rm MAP}/\chi^2_1)$. As the number of independent estimations K increases, this can be shown to approach to the interval $\hat{S}_{\rm MAP} \pm 1.96\sqrt{\hat{\kappa}_0/K}$, as predicted by the CLT.

II. ESTIMATION OF THE POWER LAW EXPONENTS AND SWITCHING RATES

The point and confidence interval estimations given int previous section can be further used to fit specific functional forms to the estimated PSDs. A rudimentary weighted least squares fitting technique has been applied to estimate the exponent r of the power-law dependencies $S(f) \sim 1/f^r$ and/or characteristic time parameter $\tau = 1/\gamma$ of the Lorentzian $S(f) \sim (1+\tau^2 f^2)^{-1}$. For simplicity we assume that the distribution of all estimated PSD points are independent Gaussians with standard deviations $(\hat{\sigma}_n)$ given by their error bars. Here $\hat{\sigma}_n$ corresponds to the PSD estimation at frequency f_n on the plots. In particular, the error bars in Fig. 2, panels c extend 1.96 standard deviations below and above the PSD point estimations $\hat{S}_{\rm MAP}(f_n)$. From the principle of maximum likelihood follows that minimizing the quantity

$$\sum_n \frac{1}{\hat{\sigma}_n^2} \left[S(f_n) - \hat{S}_{\rm MAP}(f_n)\right]^2 \tag{S11}$$

yields the 'best-fitting' $S(f)$ trial function.

Clearly, the Gaussian nature (at low frequencies) and the independence of the points are rather rough approximations. Nevertheless a simple examination of the confidence intervals in Fig 1. panels c should convince the reader that the estimated exponents can be trusted at least up to the first decimal.

III. ADDITIONAL DETAILS OF THE MRC FRAMEWORK

We recognize that there are two vastly different timescales in the system. The characteristic electronic relaxation timescale is nanoseconds, while the fluctuations (blinking) are slower then $\sim$ 100 microseconds[10]. The variation of k_n is clearly very slow compared to the timescale of electronic processes. The two timescales can be, therefore, separated.

First, consider the timescale of electronic relaxations. On this 'fast' timescale one can use the standard steady-state kinetic (rate) equations to describe the fluorescence (excitation-relaxation) cycles in the QD under steady illumination. A simple kinetic scheme is proposed. The scheme has two levels—a ground state and an excited state level—with a constant excitation rate (k_I) and a two-channel relaxation of both radiative (k_r) and non-radiative (k_n) channels (see Fig. S4). The kinetic equations are

$$\begin{aligned} \frac{dP_G}{dt} &= -k_I P_G + (k_r + k_n) P_E, \\ \frac{dP_E}{dt} &= k_I P_G - (k_r + k_n) P_E, \end{aligned} \tag{S12}$$

where P_G and P_E are the populations of the ground state and excited states, respectively, satisfying the usual normalization condition, $P_G + P_E = 1$. Due to the slow variation of k_n on the electronic relaxation timescales, its time dependence has been suppressed. Moreover, due to the above considerations, steady-state conditions are assumed to be achieved by the system.

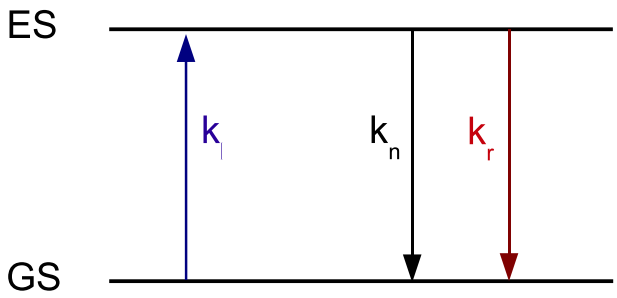


FIG. S4: Kinetic scheme for the fluorescent cycle of the MRC model. The transition rate from the ground state (GS) to excited state (ES) k_I is steadily maintained by the light excitation of the QD. Relaxation to the GS can happen via both radiative (k_r) and non-radiative (k_n). On the timescale of electronic relaxations all of these rates are practically constant.

The steady-state fluorescence emission intensity is proportional to equilibrium excited-state

population,

$$I = k_r P_E^{(0)}. \tag{S13}$$

$P_E^{(0)}$ can be calculated from Eq. (S12), by assuming $\frac{dP_G^{(0)}}{dt} = \frac{dP_E^{(0)}}{dt} = 0$, and the normalization condition $P_G^{(0)} + P_E^{(0)} = 1$. Finally, the emission intensity can be cast as

$$I = \frac{k_I k_r}{k_I + k_r + k_n}. \tag{S14}$$

Here, the excitation rate k_I is

$$k_I = \frac{IS}{\hbar\omega}, \tag{S15}$$

where I and ω are the excitation light intensity and frequency, while S is the excitation cross section[11]. For practical purposes, the k_I term from the denominator of Eq. (S14) can be neglected[12].

Secondly, considering the 'slow' timescale of the intensity fluctuation (blinking). Explicitly noting the time dependence due to the slow fluctuation in the non-radiative rate, the fluorescence intensity can be written as

$$I(t) = k_I Y(t) = \frac{k_I k_r}{k_r + k_n(t)}, \tag{S16}$$

where $Y(t)$ is the fluctuating quantum yield or quantum efficiency defined as the ratio of the number of emitted photons over the number of absorbed photons.

Note that the kinetics we have described above is an example of a quasi-static process. The idea of the fluctuating non-radiative rate was first proposed by Mews and co-authors in a study performed on the photoluminescence lifetime distribution in QDs [13]. The radiative rate of the exciton recombination k_r is assumed to be more-or-less constant, therefore the fluctuations in quantum efficiency are solely due to fluctuations in the non-radiative rate, a result observed by several other groups as well[14–16].

We now possess the relationship between the intensity and non-radiative rate. By introducing an arbitrary threshold level for the intensity I_{th}, we define the state of the QD as 'on'if $I(t) \geq I_{\text{th}}$, and 'off'if $I(t) < I_{\text{th}}$. The intensity threshold in turn corresponds to a threshold value for the non-radiative rate k_{th}. Indeed, if $k_t(t) \leq k_{\text{th}}$, the QD is 'on'and otherwise it is 'off'.

In the remainder of this section, we will show how the MRC framework can be applied to specific nano-blinkers, such as colloidal QDs, NRs and NWs by assuming model parameters in compliance with the basic physical properties of the blinking phenomenon in these systems. We will show that the number of independent parameters can be reduced to 7 (QD) or 8 (NW and NR). A list of these parameters is found at the end of the section.

Firstly, consider colloidal CdSe/ZnS QDs. In these systems the number of TLSs is typically around 10, in agreement with the experimentally obtained estimate by Scholes and coworkers [17]. We further reduce the number of parameters by assuming equal rates for trapping introducing the notation $k_i = k$, and assuming that $k_0 = 0$. The switching times have to be widely distributed in order to explain the many orders of magnitude that the fluorescence fluctuation process spans in time: from milliseconds to hours. In order to reproduce the $\sim 1/f$ PSD of the blinking trajectory, we assume that switching rates follow the geometric sequence

$$\gamma_i = \gamma_1 \, a^{i-1}. \tag{S17}$$

Note that in this case the PSD of the $k_t(t)$ fluctuation is of $1/f$ ($r = 1$) shape (see the following section). The parameter a is typically set to be $\sim 10^{-1/2}$. This guarantees that the power law covers ~ 5 orders of magnitude in time domain. The physical meaning of parameter a is related to the time-domain density of two-level systems,

$$\frac{\text{number of TLS}}{\text{decade of time}} = -\frac{1}{\log_{10} a}.$$

Typically we assume 2 TLS for each decade in time (a total of 10), enough to reproduce a smooth power law spectrum (see Fig. S5). The switching rate of the first RC (γ_1) determining the scale of time in the model is set to be $1/\tau_0$.

As reported by a number of authors there is a clustering of intensity states seen in experimental trajectories of core/shell QDs as well as fluorescent molecules leading to an approximate dichotomous ('on'-'off' type) blinking[11, 18, 19]. In order to reproduce this feature, we introduce an interaction between TLSs. The interaction has a simple mean-field form, similar to the one described in Ref. [20]. Due to the nanometer scale of the QD, an all-to-all coupling was found reasonable. Accordingly, in presence of interaction the switching rates depend on the actual state

of each TLS denoted by σ_i and are formally cast as

$$\gamma_i^{\pm} = \gamma_i \exp\left(\pm \alpha \sum_{j=1}^{N} (\sigma_j - 1/2) \pm \beta \right). \tag{S18}$$

The mean-field interaction between RCs is characterized by a strictly positive value of the parameter α that effectively results in the synchronization of the TLSs. Thus, the MRC model explains the experimentally seen bimodal distribution of intensities as a consequence of this mean-field interaction. The bias parameter β tunes the relative weights of the two intensity peaks. Note that β has to be negative to make the 'on'peak large in comparison with the 'off'peak.

Secondly, consider the possibility of a less steep PSD, i.e. when $r < 1$. Based on our PSD estimations this is the case of the NR and the "whole blinker" NW, although as discussed in the main article the decreased r could be attributed to measurement artifacts. Nevertheless, in order to explain these PSDs we have relaxed the condition imposed on the distribution of RC switching rates from Eq. (S17). We assume a $1/\gamma^r$ distribution, which is in fact a generalization of the uniform distribution in $\log(\gamma)$ from Eq. (S17). The latter corresponds to the $r = 1$ case. We control the distribution by the number of RCs (N) as well as the fastest and slowest of the switching rates ($\gamma_1$and γ_N or vice-versa). Given these 3 parameters a set of $1/\gamma^r$ distributed switching rates result if we assume that

$$\gamma_i = \left[\gamma_1^{1-r} + A(i-1) \right]^{1/(1-r)}, \quad A = \frac{\gamma_N^{1-r} - \gamma_1^{1-r}}{N-1}. \tag{S19}$$

Finally, for the colloidal CdSe/ZnS QD system the 7 independent parameters are: radiative rate (k_r), background non-radiative rate (k_0), the parameter of the TLS distribution (a), the number of TLSs (N), the switching rate of the fastest RC (γ_1), the interaction strength (α) and the interaction bias (β). The same set of parameters apply for the NR and NW system except that (1) instead of a we introduce the slowest RC switching rate, γ_N, and (2) there is an additional parameter, the power law exponent r.

IV. CALCULATING THE POWER SPECTRAL DENSITY IN THE MRC MODEL

In this section the formalism of the MRC model will be employed in order to calculate the PSD of the blinking process. Based on the results it will be shown that the framework of the MRC model is suitable to reproduce the features of the various PSDs presented in Fig. 2. We will

start from the simplest configuration in which only one ($N = 1$) RC is postulated. Following this, the calculations will be extended to the two versions proposed for the MRC model: one in which the RCs act as independent stochastic two-level systems (TLS) and another one in which there is interaction between these TLSs [21].

The system of only one RC can be in either of two states of the binary coordinate $s = \{0, 1\}$. It randomly jumps between these states according to a set of simple rate equations,

$$\frac{dp_0}{dt} = -\gamma^+ p_0 + \gamma^- p_1, \tag{S20}$$

$$\frac{dp_1}{dt} = -\gamma^- p_1 + \gamma^+ p_0, \tag{S21}$$

where $\gamma^+ = k_{2\leftarrow 1}$ and $\gamma^- = k_{1\leftarrow 2}$ denote "up" and "down" transition rates. p_0 and p_1 are time-dependent probability density functions associated with state $s = 0$ and $s = 1$, respectively. Note that a solution for these equations can be provided only if initial conditions are given first. The negative terms are due to the conservation of probability, $p_0 + p_1 = 1$. A characteristic of the TLS system is that it has an exponentially decaying correlation function. The correlation function, by definition is

$$R(t) = \langle s(t')s(t'+t)\rangle\,, \tag{S22}$$

where the notation $\langle\cdot\rangle$ stands for the usual expectation value operator. The correlation function can be expressed as a summation over all possible combination of two states and corresponding two-time probability functions. The latter can be cast as a conditional probability multiplied by a one-time probability,

$$R(t) = \sum_{s,s'} ss' P_{1|1}(s, t'+t|s', t') P_1(s', t'), \tag{S23}$$

where $P_{1|1}$ is the standard notation for conditional two-time probability. By assuming steady state conditions the one-time probability becomes time-independent, $P_1(s', t') = p_{\rm ss}(s')$. Since s and s' are defined on the domain $\{0, 1\}$, only $s = s' = 1$ terms survive in Eq. (S23). Next, the steady state probability is applied $p_{\rm ss}(1) = \gamma^+/(\gamma^+ + \gamma^-)$, and the conditional probability is expressed as the relaxation from the initially prepared $p_1 = 1, p_0 = 0$ state to the steady state with rate $\gamma^+ + \gamma^-$. Next, the correct interpretation of the kinetic equations (S20 - S21) provide us with the conditional probability $P_{1|1}(1, t'+t|1, t')$[6]. The solutions with initial conditions $p_0(t') = 0$ and $p_1(t') = 1$ yield

the needed conditional probability $p_1(t) = P_{1|1}(1, t'+t|1, t')$ [and also $p_0(t) = P_{1|1}(0, t'+t|1, t')$]. As a result, the well-known exponential form of the correlation function emerges,

$$R(t) = \left(\frac{\gamma^+}{\gamma^+ + \gamma^-} e^{-(\gamma^+ + \gamma^-)t} + \frac{\gamma^-}{\gamma^+ + \gamma^-} \right) \frac{\gamma^+}{\gamma^+ + \gamma^-}. \tag{S24}$$

From this expression we arrive to the power spectral density function of the random telegraph process by applying the Wiener-Khintchine theorem,

$$S(\omega) = \frac{1}{2\pi} \int_0^\infty R(t)(e^{i\omega t} + e^{-i\omega t}) dt \tag{S25}$$

$$= \frac{1}{\pi} \left[\frac{\gamma^+ \gamma^-}{\gamma^+ + \gamma^-} \right] \left[\frac{1}{\omega^2 + (\gamma^+ + \gamma^-)^2} \right]. \tag{S26}$$

The power spectral density thus assumes a simple Lorentzian shape with $\gamma^+ + \gamma^-$ being the HWHM (half width at half maximum) parameter. One can generalize this expression to the case when the TLS switches between any two values s_0 and $s_0 + \Delta s$ instead of 0 and 1,

$$S(\omega) = \frac{1}{\pi} \left[\frac{\gamma^+ \gamma^-}{\gamma^+ + \gamma^-} \right] \left[\frac{\Delta s^2}{\omega^2 + (\gamma_i^+ + \gamma_i^-)^2} \right]. \tag{S27}$$

Particularly, the MRC model with a single RC postulates that the two states correspond to non-radiative relaxation rates $k_n = \{k_0, k_0 + k\}$. Moreover, given the expression from Eq. (S16) the quantum yield itself also follows the same stochastic TLS kinetics with the two states in turn defined as $Y = \{k_r/(k_r + k_0), k_r/(k_r + k)\}$. Consequently both the PSD of the non-radiative rate (S_k) and quantum yield fluctuation (S_Y) follow simple Lorentzian frequency dependence, with HWHM parameter $\gamma = \gamma^+ + \gamma^-$,

$$S_k(\omega) \sim S_Y(\omega) \sim \frac{1}{\gamma^2 + \omega^2}. \tag{S28}$$

Hence, the Lorentzian PSD of the trajectory in Fig. 2.I can be explained by assuming a single RC. Similarly, a single RC dominating over all others can account for the Lorentzian feature of Fig. 2.III.

The calculation of the P_k PSD for several non-interacting RCs is straightforward. The trapping rate to the ith RC is equal to k_i in the active conformation, and zero in the passive conformation. Since the MRC model assumes that the total non-radiative rate is the algebraic sum of the

individual rates, the non-interacting RCs' independence translates into the identity

$$R_k(t) = \sum_{i=1}^{N} k_i^2 \langle \sigma_i(0)\sigma_i(t)\rangle + \frac{1}{2}\sum_{i>j>0} k_i k_j + k_0 \sum_i k_i, \tag{S29}$$

In the first term one can recognize the sum of correlations of individual rates. The remaining terms are constants that provide no contribution to the PSD. Assuming that $k_i = k = 1$, for $\omega > 0$ the PSD of the sum will be equal to the sum of individual PSDs,

$$S_k(\omega) = \frac{1}{\pi}\sum_{i=1}^{N}\left[\frac{\gamma_i^+\gamma_i^-}{\gamma_i^+ + \gamma_i^-}\right]\left[\frac{1}{\omega^2 + (\gamma_i^+ + \gamma_i^-)^2}\right]. \tag{S30}$$

One finds that about 2 RCs/decade of time is a dense enough population in order to give an inverse power law PSD. If $N = 10$, this power law extends over ~ 5 decades, as observed in experiment. Such a PSD has been plotted on Fig. S5 (blue line), together with a system of $N = 5$ TLS having only 1 TLS/decade of time (solid red line), for sake of comparison. The overlapping Lorentzians (dashed red lines) produce clearly visible ripples in the latter, while there is hardly any undulation in the former.

The calculation of the P_Y requires a formalism based on the stochastic rate or master equations treated in the full space of all possible 2^N discrete states of the N RCs [21]. With little change this formalism can be further employed in the calculation of both P_k and P_Y for the interacting MRC model as well. The master equation is

$$\frac{d\vec{p}}{dt} = \hat{W}\vec{p}, \tag{S31}$$

where $\vec{p}$ is the 2^N-dimensional probability vector and $\hat{W}$ is the stochastic matrix [6]. The steady state solution obeys the equation $\hat{W}\vec{p}_{\rm ss} = \vec{0}$. Given the formal solution of the master equation as $\vec{p}(t) = \exp(\hat{W}t)\ \vec{p}(0)$, one can further introduce the propagator [22]

$$\hat{G}(t) = \exp(\hat{W}t). \tag{S32}$$

Again, by the correct interpretation of the Master Equation (S31), the elements $G_{ij}(t)$ of this matrix give the conditional probability that the system is in state i at time t provided that it was in state j initially. The correlation function of a variable which is a function of the systems states

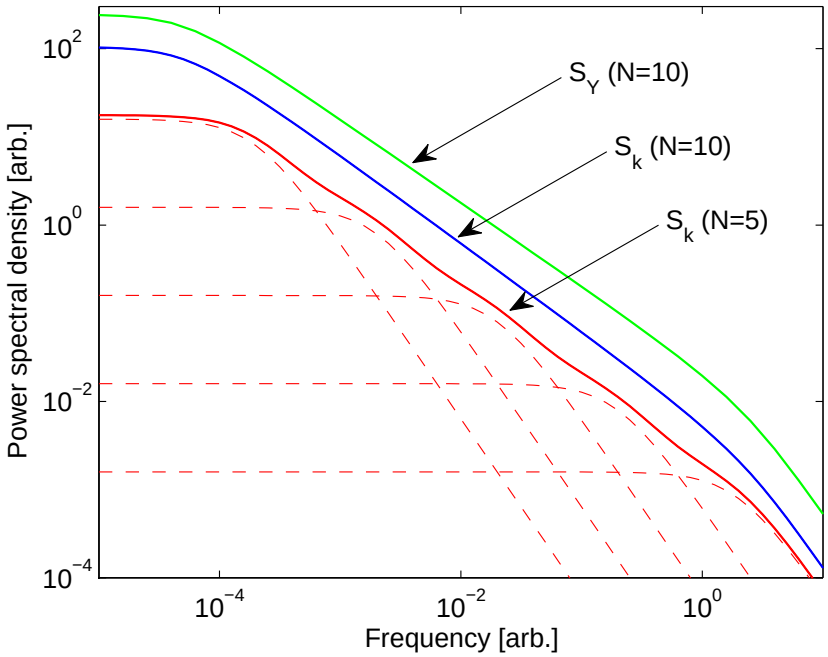


FIG. S5: Theoretical power spectral densities (PSD) calculated from the non-interacting version of the MRC model. The top (green) line is the PSD of the quantum yield fluctuation calculated for N=10. The middle (blue) and bottom (solid red) lines are the PSDs of the non-radiative rate (k) fluctuation calculated with N=10 and respectively N=5 two-level systems. The dashed curves are Lorentzians showing the input of each individual recombination center to the total fluctuation. In order to be distinguishable on a single plot the 3 PSDs have been shifted by scaling their normalization constant.

i, say Y, is by definition

$$R_Y(t) = \sum_{ij} Y_i Y_j \left\langle \vec{1}, G_{ij}(t)\vec{p}_{\rm ss} \right\rangle , \tag{S33}$$

where $\langle \cdot, \cdot \rangle$ is the scalar product operator and $\vec{1}$ is the unit vector of 2^N dimensions. We assume that the propagator can be transformed to a diagonal form, $\hat{G}(t) = \hat{V} \exp(\hat{D}t)\hat{V}^{-1}$, in terms of the eigenvalues $D_{i,i}$ and modal matrix $\hat{V}$, defined as $\hat{W} = \hat{V}\hat{D}\hat{V}^{-1}$. By applying the Wiener-Khinchine theorem from Eq. (S25), and evaluating the simple time integrals along the diagonal of $\exp(\hat{D}t)$ a formally similar expression

$$R_Y(\omega) = \sum_{ij} Y_i Y_j \left\langle \vec{1}, G_{ij}(\omega)\vec{p}_{\rm ss} \right\rangle \tag{S34}$$

follows for the PSD, with the Green's function in Fourier space being expressed for $\omega > 0$ as

$$\hat{G}(\omega) = \frac{1}{\pi} \left[\hat{V} \frac{2\hat{D}}{\hat{D}^2 + \omega^2 \hat{1}} \hat{V}^{-1} \right] . \tag{S35}$$

Note that the central term only contains diagonal matrices. In the simplest version of the interacting MRC model the non-radiative rates associated with each RC are identical, therefore the states aggregate in $N+1$ groups according to the total rate $k_n = k_0, \ldots, k_0 + Nk$. The matrix formalism described here clearly applies for aggregated states as well. The calculation of $S_Y(\omega)$ can be implemented with a suitable linear algebra package, such as the one implemented within MATLAB [23].

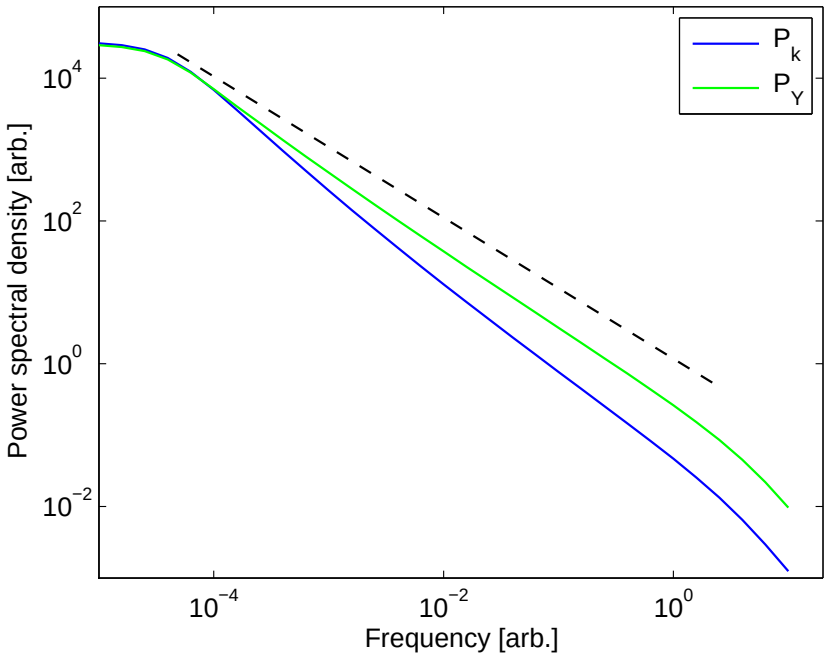


FIG. S6: Theoretical power spectral densities calculated from the interacting version of the MRC model. The interaction parameters were the same as in Ref. [21]. The bottom (blue) line is the spectrum of the non-radiative rate (k) fluctuation, while the top (green) line is the spectrum of the quantum yield fluctuation, both calculated for N=10. The dashed line is a reference: it corresponds to an inverse power law of unit exponent. In order to be distinguishable on a single plot the 3 PSDs have been shifted by scaling their normalization constant.

PSDs calculated from the interacting MRC model and plotted in Fig. S6 clearly show an inverse power law dependence $S \sim \omega^{-r}$ for both $S_k(\omega)$ and $S_Y(\omega)$, as predicted in Ref. [21]. Moreover, given the distribution of stochastic rate parameters from Eq. (S17) the non-interacting MRC resulted in an exact 1/f dependent ($r = 1$) S_k and an almost identical S_Y. However, as clearly demonstrated by Fig. S6, if interaction is turned on the exponent r for S_k becomes larger than 1 and so does the exponent for S_Y. Note that if the quantity $k_r + k_0$ is comparable to or larger then typical k_i these two exponents become approximately identical. Our calculations show that given the RC rate distribution from Eq. (S17) the power law exponents are always greater or equal to one.

The power laws found in the experimental trajectories of Fig. 2 show various exponents that nevertheless hover around one. The NR (Fig. 2, V: $r \approx 0.7$) and NW (Fig. 2, VI: $r \approx 0.9$) are special in that their exponents are below 1. This cannot be explained if Eq. (S17) holds and we have explored the possibility of the RC rate parameters being distributed as γ^{-r}. However, as mentioned in the main text, it is quite possible that the unusually low r is an artifact from experimental measurement and data preprocessing techniques.

V. SYSTEM-SPECIFIC STATISTICAL FEATURES OF FLUOROPHORE TRAJECTORIES

Self-assembled InP quantum dot (Fig. 2.I). Since this trajectory was not recorded by a photon counting device, the intensity units are not well defined and the units of the PSD are arbitrary. However, the bin time τ_0 is known and the characteristic time τ of the Lorentzian fitted to the PSD could be estimated. There is no long term correlation in this trajectory and consequently the PSD estimates are weakly correlated. The roughly 9000 intensity points allowed for usable spectral estimations spanning almost three decades in frequency. The data clearly shows a simple stochastic two-level system behavior.

The theoretical PSD in Fig. 2, I.c is a Lorentzian with corresponding to a single RC with switching rate $\gamma = 0.56 s^{-1}$.

CdSe-ZnS quantum dot (Fig. 2.II). This is an excellent trajectory: it is very long (360000 intensity points) and shows long range correlation. The PSD estimates are strongly correlated and their correlation decays in about $m_c = 100$ steps for the short windows at hight frequency. Thus it is reasonable to apply the CLT. Statistical independence between neighboring windows appears to apply for a few very long window sizes (4 points on the low frequency end). The usable spectral estimations cover 4.5 decades in frequency. We conclude that we have statistical evidence for the $1/f^r$ behavior spanning at least 4.5 decades in time.

The theoretical PSD in Fig. 2, II.c was calculated with $N = 10$ RCs with rate parameters $k_i = 10^7 s^{-1}$, $k_r = 10^7 s^{-1}$ and $k_0 = 0$, switching rates distributed homogenously in log scale in between $\gamma_N = 5 \times 10^{-3} s^{-1}$ and $\gamma_1 = 50 s^{-1}$, and interaction parameters $\alpha = 0.25$, $\beta = 0.13$. Nearly identical parameters have been assumed in our calculations from Refs. [21] and [24].

CdSe-CdS quantum dot (Fig. 2.III). With a length of about 40000 points, this trajectory is also long-correlated, similarly to that in **the previous case Fig. 2.II.** There is overwhelming statistical evidence for the Lorentzian behavior of the PSD at high frequencies. Also, a strong

evidence for $1/f^r$ spectrum is seen for the lower lower half of the frequency domain.

The theoretical PSD in Fig. 2, III.c was calculated from a total of $N = 5$ RCs with rate parameters $k_1 = 3 \times 10^6 s^{-1}$, $k_i = 6 \times 10^6 s^{-1}$, for $i > 1$, $k_r = 10^6 s^{-1}$ and $k_0 = 0$, switching rates $\gamma = 15 s^{-1}$ and the rest distributed homogenously in log scale, in the range of $\gamma_1 = 10^{-4} s^{-1}$ and $\gamma_N = 10^{-2} s^{-1}$, interaction parameters $\alpha = 0.1$, $\beta = 0$.

Alloyed CdZnSe/ZnSe quantum dot (Fig. 2.IV). For this trajectory of 40000 points the high frequency modes of the fluorescence fluctuation are deeply buried in white noise. This is the consequence of superposition of the shot noise and $1/f^r$ fluctuation. In addition to the MAP method used for all trajectories, for this particular case we found useful to apply a second method, the so-called standard Blackman-Tuckey (BT) spectral estimator. The BT method estimates the PSD from the autocorrelation function r of the process by using a lag window w [25]

$$\hat{S}_{\mathrm{BT}}(f) = \tau_0 \sum_{n=-(N-1)}^{N-1} w_n r_n \exp(i2\pi f \tau_0 n). \tag{S36}$$

The autocorrelation function shows a strong peak at the origin and a long tail. We can clearly attribute these features to the two convolved processes: the former is due to white noise, the latter is identified with the $1/f^r$ fluctuation. The shot noise was filtered out by a Wiener filter (substituting the large delta-function peak at $n = 0$ with an extrapolation of the long tail [26].)

The resulting BT estimation is plotted in Fig. 2.IV as a dotted magenta line. Clearly, a half decade (in frequency) of PSD could be deconvoluted from the noise, resulting in a total of two decades of $1/f^r$ dependence. While confidence intervals have not been calculated for the BT method, we were able to give additional evidence that supports the $1/f^r$ nature of this trajectory.

The theoretical PSD in Fig. 2, IV.c was calculated with a $N = 7$ non-interacting MRC model. The trapping rates $k_i = 10^6 s^{-1}$ were assumed to be smaller then the radiative rate, $k_r = 10^7 s^{-1}$. Further parameters were $k_0 = 0$ and the switching rates were distributed homogenously in log scale in the $\gamma_1 = 10^{-3} s^{-1}$, $\gamma_N = 3 s^{-1}$ range.

Nanorod (Fig. 2.V). These are the longest trajectories (7 hours) and among the largest in terms of data points (240000). The fluorescence fluctuation is correlated over a very long time but the saturation of $\hat{\kappa}_k^{\mathrm{cum}}(f)$ is always reached. The $1/f^r$ nature of the PSD is clearly demonstrated in this case as well.

The theoretical PSD in Fig. 2, V.c resulted from an 12 RC non-interacting model. A large non-radiative background rate is assumed, $k_0 = 10^7 s^{-1}$ relative to trapping rates $k_i = 10^6 s^{-1}$ and radiative rate $k_r = 10^6 s^{-1}$. Switching rates were power-law distributed with $r = 0.7$, and range

$\gamma_1 = 4 \times 10^{-3} s^{-1}$, $\gamma_N = 10 s^{-1}$, according to Eq. (S19).

Nanowire (Fig. 2.VI). The quantum wire trajectory has a binning times of ~ 50 ms, and the number of intensity points is a mere ~ 6500. As a result only about 2 frequency decades of PSD could be extracted. This is, however, enough to draw the conclusion that a power law dependence of the PSD applies for this trajectory as well.

The calculation of the theoretical PSD in Fig. 2, VI.c assumed an $N = 10$ non-interacting MRC model. A very large non-radiative background rate is assumed, $k_0 = 10^8 s^{-1}$ with all other rates being 100 times smaller. Switching rates were power-law distributed following Eq. (S19) with $r = 0.9$ in the interval $\gamma_1 = 0.03 s^{-1}$, $\gamma_N = 50 s^{-1}$.

Perylene diimide molecule (Fig. 2.VII). Due to the chemical structure of three aromatic planes of this molecule we see two irreversible bleaching events in the trajectory. These events could be identified with the serial bleaching of the monomers. The likely scenario is that after the two bleaching events only one plane remains in a state of high fluorescence quantum yield. The fluctuation from this monomer is analyzed, which in practice means that we discard the first transient part of the fluorescence fluctuation in our analysis. The usable length of the intensity trajectory is therefore about 22000 intensity points. From a PSD estimation point of view this trajectory was similar to the CdSe-ZnS trajectory (Fig. 2.II). The $1/f^r$ nature of the spectrum clearly applies.

The theoretical PSD in Fig. 2, VII.c was calculated with a $N = 10$ relatively strongly interacting RCs. Interaction parameters were to $r = 0.25$ and $\beta = 0$. The trapping rates $k_i = 10^6 s^{-1}$ were assumed to be smaller then the radiative rate, $k_r = 5 \times 10^6 s^{-1}$. Further parameters were $k_0 = 10^6 s^{-1}$ and the switching rates were distributed homogenously in log scale, within $\gamma_1 = 5 \times 10^{-2} s^{-1}$ and $\gamma_N = 100 s^{-1}$.

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[28] The simplest window function is the rectangular window which assigns equal weights to all points. Reasonably smooth PSD estimations were calculated for trajectories II, III, IV and VI of Fig. 2, but not for the rest of the trajectories. Thus, for consistency we settled for a typical overlap of $[M/2]$ (except for the quantum wire trajectory, Fig. 2.VI,, where we had $[0.82\,M]$) and used a Hamming window.

[29] It can be shown for 1/f processes, that the classical Fourier spectral analysis always yields a biased value for the power exponent. The bias is, however, quite small and can be neglected [27].